

EMERGING TECHNOLOGIES: AN OVERVIEW AND THEIR IMPACT

**A Comprehensive Report for Application Development and Emerging
Technologies**

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Course:
DCIT 26 – Application Development and Emerging Technologies

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I. INTRODUCTION

Technology has become an important part of everyday life. It continues to develop and affect the way people live, how people communicate, study, work, produce products, and make decisions. Some technologies are considered emerging because they are still developing, improving, and finding new applications. They can also have a significant effect on different industries and society.

Emerging technologies are not always completely new inventions. An existing technology can also become an emerging technology when it is developed further or used in a new way. Artificial intelligence, for example, was first developed many years ago but is still considered an emerging technology because it continues to progress and is being applied to different areas.

This comprehensive report will discuss the meaning of emerging technologies, their importance to an IT student, their effects on society, the difference between Augmented Reality and Virtual Reality, and the advantages and disadvantages of Artificial Intelligence.

II. OVERVIEW OF EMERGING TECHNOLOGIES

2.1 What is Emerging Technology?

Emerging technology refers to technology that is still developing and has the potential to create significant changes in society, businesses, and different industries. It does not necessarily have to be a completely new invention, but it can also describe existing technology that continues to advance or applied in new and creative ways.

According to the module 1, emerging technologies have five major characteristics, it includes: **radical novelty**, **relatively fast growth**, **coherence**, **prominent impact**, and **uncertainty and ambiguity**.

Table 1. Characteristics of Emerging Technologies

Characteristic	Explanation
Radical novelty	Technology may introduce a new way of performing a function or may give an existing technology a new application.
Relatively fast growth	Emerging technologies can experience rapid growth in users, researchers, funding, products, services, and other areas.
Coherence	Different technologies or research areas can become connected and work together toward particular purposes.
Prominent impact	Emerging technologies can create noticeable changes across industries and different areas of society.
Uncertainty and ambiguity	Their future applications and effects cannot always be predicted because the technology is still developing.

The module uses AI as an example of radical novelty because its applications can be used for different purposes. It also describes the increasing number of companies, universities, and individuals using AI as an example of relatively fast growth.

Coherence can be observed when technologies such as **cloud computing**, **artificial intelligence**, **data analytics**, and **robotics** are brought together for different uses.

Emerging technologies can also have prominent impacts. AI, for example, can affect businesses, education, healthcare, and individuals.

However, their future impact is not always certain. Some proposed applications may be successfully developed, while others may not. The possible results can also be unintended or undesirable.

III. EMERGING TECHNOLOGIES AND THEIR IMPORTANCE TO AN IT STUDENT

3.1 How Emerging Technologies Help Me as an IT Student

Emerging technologies are important to an IT student in several ways. Information Technology is a field that continuously changes. Learning about these emerging technologies equips me with the essential skills that will help me understand what is currently being developed and how these technologies can be used in real life situations.

One example that is especially important is **Artificial Intelligence**. The module describes AI as the simulation of human intelligence by machines, including learning, reasoning, and self-correction. Understanding AI gives me an idea of how computer systems can perform tasks that involve forms of learning and reasoning.

Another important example is **Machine Learning**. It is an application of AI that allows a machine to learn from experience. The module identifies possible applications of machine learning in healthcare, financial services, marketing, and transportation. As an IT student, learning about machine learning can help me understand how computer systems can process information and improve their performance.

Most importantly, studying emerging technologies encourages me to keep learning. Emerging technologies are not always finished or fully developed. Therefore, as an IT student, I cannot rely only on what has already been learned. I need to be willing to explore new tools, innovate, and understand how technology can be applied to solve problems. Studying them helps me become more prepared to adapt when new technologies and applications are introduced.

IV. EFFECTS OF EMERGING TECHNOLOGIES ON SOCIETY

Emerging technologies can create both opportunities and challenges for society. Since these technologies can have a prominent impact, their effects can reach different industries and groups of people.

4.1 Positive Effects

One positive effect is **improved efficiency**. Technologies can help people and organizations process information and perform tasks more effectively.

For example, **Big Data Analytics** involves collecting, organizing, and analyzing large amounts of data to discover patterns and useful information. It can also support real-time decision-making in production and customer management systems.

Another positive effect is the development of **healthcare applications**. Machine learning can be applied to healthcare and may assist in identifying and developing medicines more rapidly.

Emerging technologies can also improve the way devices interact. The **Internet of Things (IoT)** allows everyday physical objects to connect to the Internet and communicate with other devices. This can support real-time responses and results.

4.2 Negative Effects

Despite their benefits, emerging technologies can also create uncertainty. Since they are still developing, their future applications and consequences may not be completely known.

The module explains that some possible outcomes can be unintended or undesirable. Different groups of people may also have different interpretations and expectations regarding a technology.

Emerging technologies can also change existing industries and relationships. Industry 4.0, for example, is expected to change relationships among suppliers, producers, customers, humans, and machines.

Therefore, society should not only focus on the benefits of emerging technologies. Their possible effects and uncertainties should also be considered.

V. AUGMENTED REALITY AND VIRTUAL REALITY

5.1 What is Augmented Reality?

Augmented Reality (AR) adds computer-generated elements to a person's view of the real world. These elements can include displays, sounds, text, and effects. AR combines real-world scenes with computer-generated information to create an enhanced view.

Examples mentioned in the module include in-car displays that provide driving assistance, Pokémon GO, and applications that allow shoppers to see how furniture could look inside their homes before purchasing it.

5.2 How AR is different from VR

The module identifies both AR and VR as technologies associated with Industry 4.0, but its detailed explanation in this section focuses mainly on AR.

In simple terms, **AR enhances the real world by adding digital information**, while **VR provides a computer-generated environment for the user to experience**.

Table 2. Comparison of Augmented Reality and Virtual Reality

Augmented Reality (AR)	Virtual Reality (VR)
Adds digital elements to the real world	Creates a computer-generated environment
The real environment remains part of the experience	The experience focuses on the virtual environment
Can provide information while performing real-world activities	Can simulate an environment or experience
Example: Pokémon GO	Example: virtual simulations

Thus, the main difference is that AR adds to reality, while VR replaces the user's Surrounding with a virtual experience.

VI. ADVANTAGES AND DISADVANTAGES OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence is one of the emerging technologies discussed in the module. It involves the simulation of human intelligence processes by machines, including learning, reasoning, and self-correction.

6.1 Five advantages of Artificial Intelligence

Table 3. Advantages of AI

No.	Advantage	Explanation
1	Faster information processing	AI can process information and perform tasks involving learning and reasoning.
2	Healthcare applications	AI and machine learning can support healthcare-related research and applications.
3	Wide range of uses	AI can be applied in areas such as business, education, healthcare, and other sectors.
4	Supports decision-making	AI can be combined with data and analytics to help organizations obtain useful information.
5	Encourages innovation	Existing AI technology can be applied to different purposes and create new applications.

AI has a prominent impact because it can influence different sectors, including businesses, academic institutions, healthcare, and individuals.

6.2 Five Disadvantages of Artificial Intelligence

Table 4. Disadvantages of AI

No.	Disadvantage	Explanation
1	Uncertain outcomes	The future effects and results of emerging technologies cannot always be predicted.
2	Possible undesirable effects	Some applications may produce results that were not originally intended.
3	Uncertain future applications	Some proposed AI applications may be developed successfully while others may not.
4	Changes to industries	AI can have a significant impact that changes how industries and organizations operate.
5	Different interpretations	Different social groups may have different expectations and meanings associated with AI.

These disadvantages are connected to the module's discussion of **uncertainty and ambiguity**, which are important characteristics of emerging technologies.

VII. THE ROLE OF EMERGING TECHNOLOGIES IN INDUSTRY 4.0

The development of technology has changed the way production is performed. The module describes the industrial revolution as the systematic transformation of manufacturing through the integration of physical and digital systems to improve quality, reduce costs, and increase efficiency.

Table 5. Stages of the Industrial Revolution

Stage	Description
Industry 1.0	Introduced mechanical production using hydroelectric and steam-powered equipment.
Industry 2.0	Introduced electricity and new technological systems that enabled greater production.
Industry 3.0	Introduced electronics and information technology to further automate production.
Industry 4.0	Represents the current developing environment involving technologies such as IoT, robotics, VR, AR, and AI.

The fourth industrial revolution is particularly important because it involves the integration of several emerging technologies. These technologies are changing how people live and work and can affect relationships between humans and machines.

7.1 Big Data Analytics

Big Data Analytics involves collecting, organizing, and analyzing large amounts of data from different sources. Its purpose is to discover patterns and useful information. It can also support real-time decision-making.

7.2 Internet of Things

The Internet of Things (IoT) refers to the connection of everyday physical objects to the Internet. These devices can communicate and interact with one another and with centralized controllers. This can allow systems to respond to information in real time.

7.3 Cloud Computing

Cloud Computing refers to delivering hosted services through the Internet. According to the module, cloud technologies can support faster reaction times and allow machine data and functionality to be deployed to the cloud.

7.4 Augmented Reality in Industry 4.0

AR can also support industrial activities. The module gives examples such as selecting parts in warehouses and sending repair instructions through mobile devices. In the future, AR may provide workers with real-time information to improve decision-making and work procedures.

VIII. CONCLUSION

Emerging technologies are an important part of technological development because they can introduce new ways of solving problems and can significantly affect society. They are characterized by radical novelty, relatively fast growth, coherence, prominent impact, and uncertainty and ambiguity.

For an IT student, understanding these technologies is important because the IT industry continues to develop. AI, machine learning, IoT, cloud computing, Big Data Analytics, AR, and other technologies can provide new opportunities and applications.

Emerging technologies can have positive effects such as improving efficiency, supporting healthcare, analyzing large amounts of information, and enabling devices to communicate. However, they can also create uncertainty because their future applications and effects cannot always be predicted.

The development of these technologies is also connected to **Industry 4.0**, where physical and digital systems are increasingly integrated. Because of these developments, IT students need to remain open to learning and adapting to new technologies.

Ultimately, understanding emerging technology is not only about knowing what the technology can do. It is also about understanding its possible effects, applications, opportunities, and uncertainties. This knowledge can help future IT professionals make better decisions when using and developing technology.

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X. AI CHECKER AND GRAMMARLY RESULTS

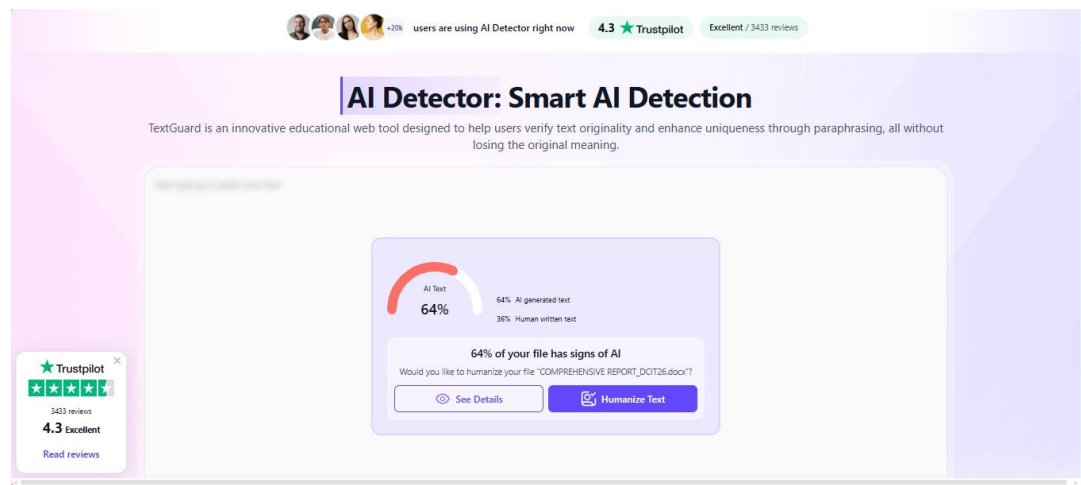
10.1 AI Checker Result

AI Checker Used: TextGuard AI

Date Checked: September 9, 2026

Result: 64% AI generated text and 36% Human written text

Screenshot:



Screenshot:

COMPREHENSIVE REPORT DCIT26

89 Overall score Share

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Grammarly

Writing quality 89

Proofread Strengthen writing

All 37 Grammar 9 Clarity 28 Delivery

Improve your text

It continues to develop and affect the way shape how people live, how people communicate, study, work, produce products, and make decisions.

Accept Dismiss

Improve wording

have a significant effect on

Improve your text

This comprehensive report will discuss the meani...

Verb problem

applied

Improve your text

It does not necessarily have to be a completely n...

Rephrase

the module

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AI Detector

Possible AI text detected 12%

Does it seem accurate?

Recommended steps

Strongest AI patterns

These sections are most likely to be detected as AI-generated.

Next

Moderate AI patterns

Weak AI patterns

Before you turn it in

AI Rewriter

Swap out phrases that may appear to be AI-generated